Celest Mech Dyn Astr (2008) 100:287–300
DOI 10.1007/s10569-008-9123-z

ORIGINAL ARTICLE

Analytic properties of Hansen coefficients

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Received: 16 October 2007 / Accepted: 5 February 2008 / Published online: 14 March 2008
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Abstract Hansen's coefficients in the theory of elliptic motion with eccentricity e are studied as functions of the parameter $\eta = (1-e^2)^{1/2}$. Their analytic behavior in the complex η plane is described and some symmetry relations are derived. In particular, for every Hansen coefficient, multiplication by suitable powers of e and η results in an entire analytic function of η. Consequently, Hansen's coefficients can be in principle computed by means of rapidly convergent series in powers of η. A representation of Hansen's coefficients in terms of two entire functions of e^2 follows.

Keywords Two-body problem · Elliptic motion · Hansen coefficients · High eccentricity · Series expansion · Analytic functions · Poles and residues

Mathematics Subject Classification (2000) 70F05

1 Introduction

This paper deals with properties of Hansen's coefficients in the theory of elliptic motion considered as functions of the parameter $\eta = \sqrt{1-e^2}$. Their expansions in powers of the eccentricity e have long been in use; comprehensive tables were compiled by Cayley in 1861 and later extended (Deprit and Rom 1967; Jarnagin et al. 1965). Further analytical developments and efficient computational techniques for expansions of various functions of elliptic motion have continued to appear, motivated by applications of such expansions in perturbation calculations (Brumberg and Fukushima 1994; Hughes 1981; Klioner et al. 1998). While classical expansions were primarily designed to handle the most important astronomical case of small eccentricities, more recent efforts have targeted the entire range $0 \le e < 1$ (Brumberg and Brumberg 1999; Klioner et al. 1998).

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We approach the problem of high eccentricities from the other end. The origin of our expansions, $\eta = 0$, corresponds to $e = 1$, a parabolic orbit.

As it turns out, the Hansen coefficients as functions of η possess nice analytic properties. As Theorem 2 (Sect. 7) asserts, every Hansen coefficient can be split into product of a simple explicit factor and an entire analytic function of η of at most exponential growth. Consequently, the second factor is represented by a series which converges uniformly in any bounded domain, in particular, in the interval $\eta \in [0, 1]$ corresponding to $e \in [0, 1]$. Moreover, the convergence is factorially fast. If desired, one can dispense with η and return to the master variable e. Breaking the entire function of η into the odd and even parts results in a decomposition $A(e^2) + \sqrt{1-e^2}\, B(e^2)$, with A, B entire functions represented by factorially convergent series (Theorem 3).

My interest to expansions centered at $\eta = 0$ comes from the study of the equation of the attitude motion of a satellite (Beletsky 1965; Chernous'ko 1964; Lutze and Abbit 1969) in the mathematically challenging limiting case $e \to 1^-$. While such a limit problem may be perceived as a purely analytical exercise, it provides an insight into classification of generalized periodic motions and their stability in the moderate, realistic eccentricity range (Bruno and Varin 2004). Higher order perturbation theory for the limit problem (Sadov 1997, 2008) requires a systematic study of expansions in η of a subfamily of Hansen's coefficients. Observations made in Sadov (2006) are generalized here.

2 Notation and basic expressions

We follow Abalakin et al. (1976) with regard to notation for orbital elements in the elliptic motion: e is the eccentricity, a the semi-major axis, M, v and E the mean, true and eccentric anomalies, r the radius vector. Introduce, as in Brumberg (1995), (2.1.9),

$$\eta = \sqrt{1-e^2},$$

and the Newcomb ratio (Abalakin et al. 1976, (3.02.21))

$$\beta = \frac{1-\eta}{e} = \frac{e}{1+\eta} = \left(\frac{1-\eta}{1+\eta}\right)^{1/2}.$$

Note that

$$e = \frac{2\beta}{1+\beta^2}, \qquad \eta = \frac{1-\beta^2}{1+\beta^2}. \tag{1}$$

We recall that r/a is determined by the ellipse equation

$$\frac{r}{a} = \frac{\eta^2}{1+e\cos v}.$$

The relation between v and M is the 2nd Kepler Law

$$dM = \eta^{-1}\left(\frac{r}{a}\right)^2 dv, \tag{2}$$

while E and v are related by

$$dE = \eta^{-1}\left(\frac{r}{a}\right) dv. \tag{3}$$

In addition to the differential Eqs. 2, 3, the normalization condition $v|_{M=0} = E|_{M=0} = 0$ is imposed.

Hansen's coefficients (Hansen 1855) $X_k^{n,m}(e)$ are defined by the Fourier series

$$\left(\frac{r}{a}\right)^n \exp(imv) = \sum_{k=-\infty}^{\infty} X_k^{n,m} \exp(ikM), \qquad i = \sqrt{-1}. \tag{4}$$

Closely related are two other sets of coefficients. Cayley's coefficients (Cayley 1861) of the real trigonometric expansions are defined as $C_0^{n,m} = X_0^{n,m}$, $C_k^{n,m} = X_k^{n,m} + X_{-k}^{n,m}$ $(k > 0)$, $S_k^{n,m} = X_k^{n,m} - X_{-k}^{n,m}$. Newcomb's coefficients (Newcomb 1895) $N_k^{n,m} = X_{k+m}^{n,m}$ appear in the Fourier expansion of $(r/a)^n \exp(im(v - M))$. Our analysis of the functions (16) in fact covers more general Andoyer coefficients (Andoyer 1923; Hagihara 1970, p.498) of the expansion of the function $(r/a)^n \exp\{i(m_1 v + m_2 E)\}$, but we do not pursue further details.

The coefficients $X_k^{n,m}$ as functions of e are represented by the Maclaurin series with real coefficients

$$X_k^{n,m} = \sum_l \hat{X}_{k,l}^{n,m} e^l, \tag{5}$$

where

$$\hat{X}_{k,l}^{n,m} = 0 \quad \text{if} \quad l < |m - k|. \tag{6}$$

This property is called the D'Alembert characteristic; it will automatically follow from our main result in Sect. 7. The coefficients $\hat{X}$ with shifted indices are known as Newcomb's operators

$$X_{\rho,\sigma}^{n,m} = \hat{X}_{m+\rho-\sigma,\,\rho+\sigma}^{n,m}.$$

Various formulas for the Hansen/Cayley/Newcomb coefficients are found, among other sources, in Abalakin et al. (1976, §3.02), Jarnagin et al. (1965), Hagihara (1970, §5.12), Tisserand (1889).

Let $s = \exp(iE)$, $t = \exp(iv)$, $z = \exp(iM)$. (This notation is consistent with Hagihara 1970, (5.118).) It follows from (2), (3) that

$$\begin{aligned} z &= s \exp\left\{\frac{e}{2}\left(s^{-1} - s\right)\right\}, \qquad t = \frac{s-\beta}{1-\beta s}, \\ \frac{r}{a} &= is\frac{dM}{ds} = -\frac{es^2 - 2s + e}{2s} = \frac{e(s-\beta)(\beta^{-1} - s)}{2s}. \end{aligned} \tag{7}$$

Hence

$$X_k^{n,m} = \left(\frac{e}{2}\right)^{n+1} \frac{\beta^{-m}}{2\pi i} \oint_{|s|=1} (s-\beta)^p (\beta^{-1} - s)^q s^r \exp\left[\frac{ke}{2}(s - s^{-1})\right] ds, \tag{8}$$

with

$$p = n + m + 1, \qquad q = n - m + 1, \qquad r = -n - k - 2. \tag{9}$$

Representation (8) goes back to Tisserand (1889, p. 253). Lyddane in Jarnagin et al. (1965, Ch. 6) used it for iterative calculations of $X_k^{n,m}$. A readily available modern reference is (Brumberg (1995), (2.3.20)).[1]

[1] In Sadov (2006) the author was unaware of the relevant references.

3 Hansen coefficients in variable η: preliminaries

Following (Sadov 2006), where the coefficients $X_k^{-3,2}$ (named the Chernousko functions Φ_{k-2}) are treated in detail, set

$$s = \beta x. \tag{10}$$

Define the functions (cf. relations (1))

$$\begin{aligned} \lambda(x,\eta) &= e(s^{-1} - s) = \eta(x + x^{-1}) + (x^{-1} - x), \\ \mu(x,\eta) &= e(\beta^{-1} - s) = \eta(1 + x) + (1 - x). \end{aligned} \tag{11}$$

Let $p, q, r \in \mathbb{Z}$ and $\tau \in \mathbb{C}$ be independent parameters. Introduce

$$f_{r,\tau}^{p,q}(x,\eta) = (x-1)^p\, x^r\, \mu(x,\eta)^q \exp\Big[-\frac{\tau}{2}\lambda(x,\eta)\Big]. \tag{12}$$

Denote

$$\gamma(a,b) = (1+\eta)^a (1-\eta)^b, \tag{13}$$

so that $e = \gamma(\frac{1}{2}, \frac{1}{2})$ and $\beta = \gamma(-\frac{1}{2}, \frac{1}{2})$. For reference, note the properties

$$\gamma(a,b)^{-1} = \gamma(-a,-b), \qquad \gamma(a,b)|_{\eta\mapsto-\eta} = \gamma(b,a). \tag{14}$$

Set

$$\xi(\eta) = \beta^{-2} = \gamma(1,-1). \tag{15}$$

The functions (12) are analytic in x with possible singularities at the points $x = 0$, $x = 1$ and $x = \xi$. The point $x = 0$ is essential singularity (unless $\tau = 0$ or $\eta = -1$). The point $x = 1$ is a pole of order $|p|$ if $p < 0$, and the point $x = \xi$ is a pole of order $|q|$ if $q < 0$. Let C be a simple contour enclosing the points $x = 0$ and $x = 1$ and leaving $x = \xi$ outside. Such a contour exists if $\xi \neq 0$ and $\xi \neq 1$. Two exceptional cases are: $\xi = 0$ corresponding to $\eta = -1$, and $\xi = 1$ corresponding to $\eta = 0$.

Define (assuming positive direction of integration unless otherwise noted)

$$F_{r,\tau}^{p,q}(\eta) = \frac{1}{2\pi i}\oint_C f_{r,\tau}^{p,q}(x,\eta)\,dx. \tag{16}$$

Equations (8)–(11) yield

$$X_k^{n,m} = \frac{e^m \beta^{-k}}{2^{n+1}}\, F_{r,k}^{p,q}(\eta) = \frac{\gamma(\frac{m+k}{2}, \frac{m-k}{2})}{2^{n+1}}\, F_{r,k}^{p,q}(\eta). \tag{17}$$

For further analysis we introduce, for the residues of $f_{r,\tau}^{p,q}$ at $x = \sigma$, where $\sigma = 0, 1, \xi$, or ∞, the notation

$${}_\sigma F_{r,\tau}^{p,q}(\eta) = \operatorname*{Res}_{x=\sigma} f_{r,\tau}^{p,q}(x,\eta). \tag{18}$$

The residue at infinity is understood as the corresponding Cauchy integral over a *negatively* oriented big circle. By the residue theorem, for $\eta \neq 0, -1$

$$F_{r,\tau}^{p,q}(\eta) = {}_0F_{r,\tau}^{p,q}(\eta) + {}_1F_{r,\tau}^{p,q}(\eta) = -{}_\xi F_{r,\tau}^{p,q}(\eta) - {}_\infty F_{r,\tau}^{p,q}(\eta). \tag{19}$$

The integrand in (16) depends analytically on η as long as the point $x = \xi$ can be avoided by the contour C under small variations of η. This last condition is fulfilled provided that $\eta \neq 0, -1$. Summarizing, we state

Proposition 1 *The Hansen coefficient* $X_k^{n,m}$ *admits factorization* (17)*, where* p, q, r *are defined in* (9) *and* $F_{r,k}^{p,q}(\eta)$ *is an analytic function in* $\mathbb{C}\backslash\{0, -1\}$.

4 Symmetry relations

The ingredients of the function (12) possess the following transformation properties under two involutive changes of variable, $x = \xi/x_1$ and $x = 1/x_2$:

$$\begin{aligned} \lambda(x,\eta) &= -\lambda(x_1,\eta) = -\lambda(x_2,-\eta), \\ \mu(x,\eta) &= (1+\eta)x_1^{-1}(x_1-1) = x_2^{-1}\mu(x_2,-\eta), \\ x-1 &= (1-\eta)^{-1}x_1^{-1}\mu(x_1,\eta) = -x_2^{-1}(x_2-1). \end{aligned} \tag{20}$$

In addition, $dx = -\xi x_1^{-2}\,dx_1 = -x_2^{-2}\,dx_2$. We have

$$f_{r,\tau}^{p,q}(x,\eta)\,dx = -\gamma(q+r+1,-p-r-1)\,f_{-p-q-r-2,-\tau}^{q,p}(x_1,\eta)\,dx_1, \tag{21}$$

where γ is defined in (13). Using the notation

$$\tilde{p} = q, \quad \tilde{q} = p, \quad \tilde{r} = -p-q-r-2, \quad \tilde{\tau} = -\tau, \tag{22}$$

the transformation (21) can be written in a symmetric form

$$\gamma\left(\frac{p-r}{2}, \frac{r-q}{2}\right) f_{r,\tau}^{p,q}(x,\eta)\,dx = -\gamma\left(\frac{\tilde{p}-\tilde{r}}{2}, \frac{\tilde{r}-\tilde{q}}{2}\right) f_{\tilde{r},\tilde{\tau}}^{\tilde{p},\tilde{q}}(x_1,\eta)\,dx_1.$$

Similarly, with the change of variable $x \mapsto x_2$ we get

$$f_{r,\tau}^{p,q}(x,\eta)\,dx = (-1)^p f_{\tilde{r},\tilde{\tau}}^{p,q}(x_2,-\eta)\,dx_2. \tag{23}$$

The considerations based on the involutions $\psi_1 : x \mapsto x_1$ and $\psi_2 : x \mapsto x_2$ lead to interesting and useful identities.

Proposition 2 (Transposition of p and q)

(a) *The residues* (18) *satisfy the relation*

$$_{\sigma}F_{r,\tau}^{p,q}(\eta) = -\gamma(q+r+1,-p-r-1)\;{}_{\sigma_1}F_{\tilde{r},\tilde{\tau}}^{q,p}(\eta), \tag{24}$$

where $\sigma_1 = \psi_1(\sigma)$*, that is, the (unordered) pair* $\{\sigma, \sigma_1\}$ *is either* $\{1, \xi\}$ *or* $\{0, \infty\}$.

(b) *The functions* $F_{r,\tau}^{p,q}$ (16) *satisfy the symmetry relation*

$$F_{r,\tau}^{p,q}(\eta) = \gamma(q+r+1,\,-p-r-1)\,F_{\tilde{r},\tilde{\tau}}^{\tilde{p},\tilde{q}}(\eta). \tag{25}$$

Proof The map ψ_1: $x \mapsto x_1$ transposes $1 \leftrightarrow \xi$ and $0 \leftrightarrow \infty$. A small contour enclosing $x = 1$ is carried to a likewise oriented small contour enclosing $x_1 = \xi$. A small contour enclosing $x = 0$ is carried to an oppositely oriented contour enclosing the points 0, 1 and ξ in the x_1-plane. Recalling a comment made after (18) about the sign of the residue at infinity and using (21), we obtain (24). Application of (24) to (19) yields (25). □

Proposition 3 (Changing sign of η)

(a) *The residues* (18) *satisfy the relation*

$$
\begin{aligned}
{}_0F^{p,q}_{r,\tau}(\eta) &= (-1)^p\ {}_\infty F^{p,q}_{\tilde r,\tilde\tau}(-\eta),\\
{}_\sigma F^{p,q}_{r,\tau}(\eta) &= (-1)^p\ {}_\sigma F^{p,q}_{\tilde r,\tilde\tau}(-\eta), \qquad \sigma = 1, \xi.
\end{aligned}
\tag{26}
$$

(b) *The function $F^{p,q}_{\cdot\cdot}(\eta)$ can be expressed in terms of the values at $-\eta$ of another function (with different lower indices) $F^{p,q}_{\cdot\cdot}$ and the corresponding residue functions:*

$$
F^{p,q}_{r,\tau}(\eta) = (-1)^{p+1} F^{p,q}_{\tilde r,\tilde\tau}(-\eta) + (-1)^p \left({}_1F^{p,q}_{\tilde r,\tilde\tau}(-\eta) \ - \ {}_\xi F^{p,q}_{\tilde r,\tilde\tau}(-\eta) \right). \tag{27}
$$

Proof Consider the transformations induced by the map $\psi_2 : x \mapsto x_2 = x^{-1}$, see the right column in (20). Correspondence of the σ values $0 \leftrightarrow \infty$ is analyzed in exactly same way as in the previous proof and leads to the first line in (26). The second line with $\sigma = 1$ follows from the fixed point property $\psi_2(1) = 1$. The case $\sigma = \xi$ needs more attention due to a notational subtlety. The symbol ξ in the notation ${}_\xi F$ should be considered as a label (index) attached to the pole at $x = \xi(\eta)$, which moves as η changes, and not as a fixed complex number. Specifically, as a result of the change of variable $x \mapsto x_2$ in the function $f^{p,q}_{r,\tau}$, a pole at $x = \xi(\eta)$ (if there is one) moves to the new location $x = \xi(-\eta)$. Denote $\psi_2^*(\eta) = -\eta$; then ξ as a function of η has the transformation property

$$
\xi(\psi_2^*(\eta)) = \xi(-\eta) = [\xi(\eta)]^{-1} = \psi_2\,(\xi(\eta)).
$$

Thus ${}_\xi F(-\eta)$ should be understood as $\mathrm{Res}_{x=\psi_2(\xi)}\, f(x, \psi_2^*(\eta))$ and not as $\mathrm{Res}_{x=\xi}\, f(x, \psi_2^*(\eta))$. With this in mind, the case $\sigma = \xi$ in (26) follows from (20) like the other cases.

The identity (27) follows from (19) and (26). □

Remarks

1. Hansen's functions $X^{n,m}_k$ depend on three integer indices. The functions F have four independent parameters p, q, r, τ. The reduction (17) of $X^{\cdot\cdot}_{\cdot}$ to $F^{\cdot\cdot}_{\cdot\cdot}$ imposes a linear constraint on the values of p, q, r, and $\tau = k$; from (9) we see

$$
p + q + 2(r + \tau + 1) = 0. \tag{28}
$$

   This constraint is preserved by the transformation (22).
2. When combining the transformation (26) and (27) with (24) and (25) one should carefully watch the indices and the arguments of the functions F and use the properties (14) of the symbol γ.

5 Functions $F^{p,q}_{r,\tau}(\eta)$ are meromorphic

According to Proposition 1, the only singular points of the functions $F^{p,q}_{r,\tau}(\eta)$ may occur at $\eta = 0$ and $\eta = -1$. Using the residue decomposition (19), we will show that singularities of $F^{p,q}_{r,\tau}(\eta)$, if they are present, are poles, that is, all these functions are meromorphic.

Analysis of the point $\eta = 0$

For every x with $|x| = \varepsilon \ll 1$ the function $f_{r,\tau}^{p,q}(x,\eta)$ has a finite limit as $\eta \to 0$. Therefore, the residue function ${}_0F_{r,\tau}^{p,q}(\eta)$ is regular at $\eta = 0$. Thus, a possible singularity of $F_{r,\tau}^{p,q}(\eta)$ at $\eta = 0$ can only be contributed by ${}_1F_{r,\tau}^{p,q}(\eta)$. If $p \geq 0$, then the integrand (12) has no pole at $x = 1$, hence ${}_1F_{r,\tau}^{p,q} \equiv 0$. The next proposition covers the case $p < 0$.

Proposition 4

(a) *If $p < 0$, $q \geq 0$, then*

$$ {}_1F_{r,\tau}^{p,q}(\eta) = P_{q+\lfloor \frac{|p|-1}{2} \rfloor}(\eta)\, exp(-\tau\eta), $$

where P_N denotes a polynomial of degree at most N and $\lfloor\cdot\rfloor$ is the integral part function.

(b) *If $p < 0$, $q < 0$, then*

$$ {}_1G_{r,\tau}^{p,q}(\eta) = \eta^{|q|+|p|-1}\, {}_1F_{r,\tau}^{p,q}(\eta) $$

is an entire analytic function of the form $P_N(\eta)exp(-\tau\eta)$, $N = \lfloor \frac{3}{2}(|p|-1) \rfloor$, and

$$ {}_1G_{r,\tau}^{p,q}(0) = \frac{(|q|+|p|-2)!}{(|p|-1)!(|q|-1)!}\, 2^{1-|q|-|p|}. \tag{29} $$

In both cases the coefficients of the polynomial P_N depend on p, q, r and τ.

Proof By Cauchy's formula, the dependence of ${}_1F$ on η comes from x-derivatives of orders $\leq |p|-1$ at $x = 1$ of the functions $\exp(-\tau\lambda(x,\eta)/2)$ and $\mu(x,\eta)^q$ in the integrand (12). Since $\mu(x,\eta)$ is a linear function of x, we have

$$ \left(\frac{d}{dx}\right)^{\alpha}\Bigg|_{x=1} \mu(x,\eta)^q = q(q-1)\ldots(q-\alpha+1)(2\eta)^{q-\alpha}(\eta-1)^{\alpha}. \tag{30} $$

Taking into account the identities

$$ \lambda(1,\eta) = \mu(1,\eta) = 2\eta, \tag{31} $$

an easy check by induction shows that

$$ \left(\frac{d}{dx}\right)^{\alpha}\Bigg|_{x=1} \exp\left[-\frac{\tau}{2}\lambda(x,\eta)\right] = P_{\lfloor \frac{\alpha}{2} \rfloor}(\eta)\exp(-\tau\eta). $$

If $q \geq 0$, then (30) is a polynomial of degree at most q for any α and part (a) of the Proposition follows.

In the case $p < 0, q < 0$ the right-hand side of (30) has a pole of order $|q| + \alpha$ at $\eta = 0$. The most singular term in ${}_1F$ at $\eta = 0$ occurs when all $|p| - 1$ derivatives with respect to x of the product (12) fall on the factor $\mu(x,\eta)^q$, that is, when $\alpha = |p| - 1$ in (30). Hence the order of the pole: $|q| + |p| - 1$.

Formula (29) follows from (30) and the evaluations (31). □

Analysis of the point $\eta = -1$

It is convenient to study the behavior of $F_{r,\tau}^{p,q}(\eta)$ for η close to -1 by means of the second decomposition (17), $F = -{}_\xi F - {}_\infty F$. For every x with $|x| = R \gg 1$ the function $f_{r,\tau}^{p,q}(x,\eta)$ has a finite limit as $\eta \to -1$. Therefore, the residue function ${}_\infty F_{r,\tau}^{p,q}(\eta)$ is regular at $\eta = -1$. Thus, a possible singularity of $F_{r,\tau}^{p,q}(\eta)$ at $\eta = -1$ can only be contributed by the term

$-_\xi F^{p,q}_{r,\tau}(\eta)$. If $q \geq 0$, then the integrand (12) has no pole at $x = \xi$, hence ${}_\xi F^{p,q}_{r,\tau} \equiv 0$. In the case $q < 0$ we use the relation (24) to express ${}_\xi F$ in terms of ${}_1F$:

$$-_\xi F^{p,q}_{r,\tau}(\eta) = \gamma(q + r + 1, -p - r - 1)\ {}_1F^{q,p}_{\tilde{r},\tilde{\tau}}(\eta). \tag{32}$$

The factor ${}_1F$ is regular at $\eta = -1$. The first factor, $\gamma(\ldots)$, is regular at $\eta = -1$ if $q+r+1 \geq 0$. In the case $q + r + 1 < 0$ the function (32) has pole at $\eta = -1$ of order at most $-(q + r + 1)$.

The order is exactly equal to $-(q + r + 1)$ if and only if ${}_1F^{q,p}_{\tilde{r},\tilde{\tau}}(-1) \neq 0$. By (11), (12) and (22)

$$f^{q,p}_{\tilde{r},\tilde{\tau}}(x, -1) = (-2)^p (x - 1)^q x^{\tilde{r}+p} \exp(\tilde{\tau} x) = (-2)^p (x - 1)^q x^{-q-r-2} \exp(-\tau x).$$

The condition ${}_1F^{q,p}_{\tilde{r},\tilde{\tau}}(-1) \neq 0$ is equivalent (for $q < 0$ and $q + r + 1 < 0$) to

$$\left(\frac{d}{dx}\right)^{|q|-1} \left(x^{-q-r-2} \exp(-\tau x)\right)\Big|_{x=1} \neq 0, \tag{33}$$

which holds true in general position (in the absence of a certain relation between the parameters q, r and τ).

A similar application of the relation (25) estimates the order of possible zero of $F^{p,q}_{r,\tau}(\eta)$ at $\eta = 1$. The function $F^{q,p}_{\tilde{r},\tilde{\tau}}(\eta)$ is regular at $\eta = 1$. Hence if $p + r + 1 < 0$, then $F^{p,q}_{r,\tau}(\eta)$ has zero of order at least $-(p + r + 1)$ at $\eta = 1$.

Combining the above findings and noting that the value (29) is nonzero, we obtain

Proposition 5 *The functions $F^{p,q}_{r,\tau}(\eta)$ are meromorphic with only possible poles at $\eta = 0$ and $\eta = -1$. More precisely, let*

$$\kappa_0 = \begin{cases} -p - q - 1, & \text{if} \quad p < 0, q < 0 \\ 0, & \text{otherwise} \end{cases} \tag{34}$$

and

$$\kappa_1 = \begin{cases} -q - r - 1, & \text{if} \quad q < \min(0, -r - 1) \\ 0, & \text{otherwise} \end{cases} \tag{35}$$

The function $\hat{F}^{p,q}_{r,\tau}(\eta) = \eta^{\kappa_0}(\eta + 1)^{\kappa_1} F^{p,q}_{r,\tau}(\eta)$ is an entire analytic function. If $\kappa_0 > 0$, then always $\hat{F}^{p,q}_{r,\tau}(0) \neq 0$. If $\kappa_1 > 0$, then $\hat{F}^{p,q}_{r,\tau}(-1) \neq 0$ under condition (33). Finally, if $p + r + 1 < 0$, then $(d/d\eta)^j \hat{F}^{p,q}_{r,\tau}(1) = 0$ for $j = 0, \ldots, -(p + r + 2)$.

6 Local behavior of Hansen coefficients at $\eta = 0, \pm 1$

Recall the notion of the order of an analytic function at a point (Ahlfors 1966, §4.3.3). If a function $g(\eta)$ is analytic in a punctured neighborhood of $\eta_0 \in \mathbb{C}$, then the order of g at η_0 is the integer (if it exists)

$$\operatorname{ord}_{\eta_0} g = \max\{N \in \mathbb{Z} : \ g(\eta) = O(\eta - \eta_0)^N \text{ as } \eta \to \eta_0\}.$$

The order is negative at poles and positive at zeros of the function. If η_0 is an essential singular point for g, then by definition $\operatorname{ord}_{\eta_0} g = -\infty$. The function g is meromorphic if $\operatorname{ord}_\eta g > -\infty$ for all $\eta \in \mathbb{C}$.

The notion of order directly extends to functions with algebraic branch points, like $(\eta - \eta_0)^{1/2}$. The order at a branch point will be fractional.

In particular (see (13)), for any rational a and b

$$\operatorname{ord}_{\eta=-1} \gamma(a,b) = a, \qquad \operatorname{ord}_{\eta=1} \gamma(a,b) = b. \tag{36}$$

Theorem 1 *If $m-k$ is even, then $X_k^{n,m}(\eta)$ is a meromorphic function. If $m-k$ is odd, then $(1-\eta^2)^{1/2} X_k^{n,m}(\eta)$ is meromorphic. The orders of the function $X_k^{n,m}(\eta)$ at $\eta = 0, \pm 1$ are described as follows:*

$$\operatorname{ord}_{\eta=0} X_k^{n,m}(\eta) \begin{cases} = 2n+3 < 0, & \text{if } n+1+|m| < 0, \\ \geq 0, & \text{if } n+1+|m| \geq 0. \end{cases} \tag{37}$$

$$\operatorname{ord}_{\eta=1} X_k^{n,m}(\eta) \geq \frac{|m-k|}{2}, \tag{38}$$

$$\operatorname{ord}_{\eta=-1} X_k^{n,m}(\eta) \geq \begin{cases} (m+k)/2, & \text{if } -k < m \leq n+1, \\ -|m+k|/2, & \text{otherwise.} \end{cases} \tag{39}$$

Proof By Proposition 1, the function $X_k^{n,m}(\eta) e^{(m-k) \bmod 2}$ is a single-valued analytic function in $\mathbb{C} \backslash \{0, -1\}$. It is meromorphic by Proposition 5.

To prove (37)–(39), we simply need to interpret Proposition 5 in terms of the functions $X_k^{n,m}$ using (9), (17) and (36).

First, $\operatorname{ord}_0 X_k^{n,m} = \operatorname{ord}_0 F_{r,k}^{p,q}$. The condition $p < 0, q < 0$ translates into $n+1+|m| < 0$, and the order at 0 corresponding to this condition is $-\kappa_0 = p+q+1 = 2n+3$. This proves the first line in (37). In the case $n+1+|m| \geq 0$, Proposition 5 implies that $F_{r,k}^{p,q}$ is analytic at $\eta = 0$, hence the second line in (37).

Next, $\operatorname{ord}_1 X_k^{n,m} = \operatorname{ord}_1 F_{r,k}^{p,q} + \frac{m-k}{2}$. The condition $p+r+1 < 0$ translates into $m-k < 0$. If this is true, then by the last part of Proposition 5 $\operatorname{ord}_1 X_k^{n,m} \geq -(m-k) + \frac{m-k}{2} = \frac{|m-k|}{2}$. If $m-k \geq 0$, then $\operatorname{ord}_1 X_k^{n,m} \geq 0 + \frac{m-k}{2} = \frac{|m-k|}{2}$. Thus (38) holds true.

Finally, $\operatorname{ord}_{-1} X_k^{n,m} = \operatorname{ord}_{-1} F_{r,k}^{p,q} + \frac{m+k}{2}$. The condition in (35), $q < \min(0, -r-1)$, translates into $m > \max(n+1, -k)$. If this is true, then $\operatorname{ord}_{-1} X_k^{n,m} \geq -\kappa_1 + \frac{m+k}{2} = (-m-k) + \frac{m+k}{2} = \frac{-|m+k|}{2}$. If $m \leq \max(-k, n+1)$, then $\operatorname{ord}_{-1} X_k^{n,m} \geq 0 + \frac{m+k}{2}$. In the subcase $n+1 \geq m > -k$ we have $m+k > 0$. This completes the proof of (39) and the Theorem. □

7 Global analytic properties of Hansen coefficients

Let $\kappa_0 = \kappa_0(m,n) = -2n-3$, if $n+1+|m| < 0$; otherwise let $\kappa_0 = 0$. This definition is consistent with (34) upon substitutions (9).

Theorem 2 *The function*

$$Y_k^{n,m}(\eta) = e^{|m+k|} \eta^{\kappa_0} X_k^{n,m}(\eta)$$

extends from the interval $(0,1)$ *to an entire analytic function of complex variable η. Its growth as $|\eta| \to \infty$ is estimated by the inequality*

$$|Y_k^{n,m}(\eta)| \leq C(\varepsilon) exp\{(|k|+\varepsilon)|\eta|\} \tag{40}$$

for any $\varepsilon > 0$ and some constant $C(\varepsilon)$ (possibly depending also on n, m, k).

Proof According to (37), $\mathrm{ord}_{\eta=0} X_k^{n,m}(\eta) \geq -\kappa_0(m,n)$ for all m, n, k. Theorem 1 together with obvious formula $\mathrm{ord}_{\eta=\pm 1}\, e = 1/2$ imply the estimates $\mathrm{ord}_{\eta=0,\pm 1}\, Y_k^{n,m}(\eta) \geq 0$. Hence $Y_k^{n,m}$ has no poles; so it is entire.

To estimate the growth rate of $Y_k^{n,m}$, notice that $\xi(\eta) \to -1$ as $|\eta| \to \infty$. Therefore, if $\varepsilon > 0$ is fixed and small, then for all $\eta \in \mathbb{C}$ with sufficiently large $|\eta|$ the contour $C = \{|x - \varepsilon| = 1\}$ encloses the point $x = 1$ but not $x = \xi(\eta)$. So the integral representation (16) with such a contour is correct. The obvious inequality $|\lambda(x,\eta)| \leq 2|x|(|\eta| + 1)$ implies

$$|F_{r,k}^{p,q}(\eta)| = O\left(|\eta|^N \exp\{(1+\varepsilon)|k\eta|\}\right)$$

with some positive N independent of η but possibly depending on p, q, r and k. This estimate shows that growth of $F_{r,k}^{p,q}$ is dominated by $\exp((|k| + \varepsilon)|\eta|)$ as $|\eta| \to \infty$, where ε can be made as small as we wish. It remains to note that $Y_k^{n,m}$ is $F_{r,k}^{p,q}$ times a rational function of η; the rational factor does not affect the exponential estimates. A factor of the form $O(|\eta|^N)$ is not included in (40), since it can be removed by choosing a slightly greater value of ε. □

Corollary *Coefficients in the Maclaurin series of the functions $Y_k^{n,m}(\eta)$ in Theorem 2,*

$$Y_k^{n,m}(\eta) = \sum_{j=0}^{\infty} c_{k,j}^{n,m} \eta^j, \tag{41}$$

satisfy the estimate

$$|c_{k,j}^{n,m}| \leq C(\varepsilon) \frac{(|k| + \varepsilon)^j}{j!}, \tag{42}$$

where $\varepsilon > 0$ can be arbitrarily small and $C(\varepsilon)$ may depend on n, m, k but is independent of j.

Proof By (40), the integral of $\eta^{-j-1} Y_k^{n,m}(\eta)$ over the circle $|\eta| = R$ has an upper bound $CR^{-j}\exp(AR)$, where $A = |k| + \varepsilon$. The minimum of this bound (attained at $R = j/A$) is $A^j j^{-j} \exp j = O(A^j \sqrt{j}/j!)$ by Stirling's formula. The factor $\sqrt{j}$ is removed by choosing a slightly greater value of ε. □

Theorem 3 *By the process of analytic continuation the Hansen coefficient $X_k^{n,m}$ as a function of the eccentricity e extends from the interval $[0, 1]$ to the Riemann surface of the function $\eta(e) = \sqrt{1 - e^2}$. It admits a global representation involving two entire analytic functions $A(e^2)$ and $B(e^2)$ depending on n, m and k:*

$$X_k^{n,m}(e) = e^{-|m+k|}(1 - e^2)^{-\kappa_0} \left(A(e^2) + \sqrt{1 - e^2}\, B(e^2)\right), \tag{43}$$

with κ_0 the same as in Theorem 2. The functions $A(w)$ and $B(w)$ satisfy the growth estimate as $|w| \to \infty$

$$\max(|A(w)|, |B(w)|) \leq C(\varepsilon) \exp\{(|k/2| + \varepsilon)|w|\}. \tag{44}$$

Proof The entire function $Y_k^{n,m}(\eta)$ in Theorem 2 can be written as the sum $A_1(\eta^2) + \eta B_1(\eta^2)$ with entire functions A_1 and B_1. It remains to set $A(u) = A_1(1-u)$ and $B(u) = B_1(1-u)$. The estimate (44) follows from Theorem 2 and the asymptotic equivalence $|\eta| \sim |e|$ when $|\eta| \gg 1$. □

Remarks

1. The inequality (44) implies factorial decay estimates analogous to (42) for Maclaurin coefficients of the functions A and B in powers of their argument $w = e^2$.
2. The point $e = 0$ is a regular point on the Riemann surface of the function $\eta(e)$. On the "physical sheet" the value of η corresponding to $e = 0$ is 1. Hence the order of the function $X_k^{n,m}(e)$ at $e = 0$ on the physical sheet is equal to $\mathrm{ord}_{\eta=1} X_k^{n,m}(\eta) \geq |m-k|/2$ as stated in Theorem 1. Thus the D'Alembert characteristic (6) is naturally incorporated in our theory. Interestingly, however, the order of $X_k^{n,m}(e)$ at $e = 0$ on the non-physical sheet (where $\eta = -1$) can be less than $|m-k|/2$; it may even be negative.

8 Examples

Coefficients in the series (41) or expansions of the functions $A(e^2)$, $B(e^2)$ of Theorem 3 are not easily available in closed form. In practice it may be easier to derive representations in another form, from which power expansions can be obtained by re-development.

Example 1 Consider the well-known hypergeometric series representation for $X_0^{n,m}$ (Brumberg 1995, (2.3.33))

$$X_0^{n,m} = \frac{(-n-|m|-1)_{|m|}}{|m|!}\left(\frac{e}{2}\right)^{|m|} F\left(\frac{|m|-n-1}{2}, \frac{|m|-n}{2}, 1+|m|;\ e^2\right).$$

The series is terminating if $n \geq |m|-1$, so the function $X_0^{n,m} e^{-|m|}$ is a polynomial (in e^2 or in η^2). This fact agrees with Theorem 2, but the exponent $|m|$ of the regularizing factor $e^{|m|}$ in Theorem 2 is not optimal in this case.

In another hypergeometric representation (Brumberg 1995, (2.3.34))

$$X_0^{n,m} = \frac{(-n-|m|-1)_{|m|}}{|m|!}\left(\frac{e}{2}\right)^{|m|} \eta^{2n+3}$$

$$\times F\left(\frac{|m|+n+2}{2}, \frac{|m|+n+3}{2}, 1+|m|;\ e^2\right)$$

the series terminates if $n+1+|m| < 0$ and the function $X_0^{n,m} e^{-|m|}\eta^{-2n-3}$ is a polynomial. The exponent of η here is as that in Theorem 3 (cf. (37)), while the exponent of e in Theorem 3 is, again, not optimal.

If $-|m| \leq n+1 < 0$, then $X_0^{n,m} \equiv 0$. In the remaining range $0 \leq n+1 < |m|$, the most convenient representation is (Brumberg 1995, (2.3.31))

$$X_0^{n,m} = \frac{(-n-|m|-1)_{|m|}}{|m|!}\beta^{|m|}(1+\beta^2)^{-n-1}$$

$$\times F\left(|m|-n-1,\ -n-1,\ 1+|m|;\ \beta^2\right).$$

The series here is a polynomial of degree $n+1$ in $\beta^2 = (1-\eta)(1+\eta)^{-1}$. An easy direct estimation of orders at $\eta = \pm 1$ gives: $\mathrm{ord}_{\eta=1} X_0^{n,m} \geq |m|$, $\mathrm{ord}_{\eta=-1} X_0^{n,m} \geq -|m|+n+1-(n+1) = -|m|$. These estimates coincide with those given by Theorem 2. Now the power factor $e^{|m|}$ in Theorem 3 required to kill the pole at $\eta = -1$ cannot be reduced.

Example 2 A detailed analysis of the functions $F^{0,-4}_{1-k,k}$ corresponding to Hansen's coefficients $X^{-3,2}_k$ is carried out in Sadov (2006). In particular, it is shown that for $k > 0$

$$F^{0,-4}_{1-k,k}(\eta) = \frac{(1-\eta)^{k-2}}{(\eta+1)^{k+2}} \exp(k\eta) \sum_{j=0}^{\infty} \frac{a_j}{j!} \eta^j. \tag{45}$$

The power series here represents an entire function (denoted $F_k(\eta)$ in Sadov (2006)). For $k = 0$ and $k = 1$ the singularity of the prefactor $(1-\eta)^{k-2}$ at $\eta = 1$ is compensated by zero of the corresponding order of the power series (Sadov 2006, (41)), in agreement with Proposition 1. The first few coefficients of the series are (Sadov 2006, Theorem 5a)

$$a_0 = \frac{k}{12}\left(-1 + J_0(k) + \int_0^k J_0(x)\,dx - 2\sum_{l=0}^{k-1} J_l(k) - J_{k-1}(k)\right),$$
$$a_1 = \frac{k^2}{4}(J_{k-1}(k) - J_k(k)), \qquad a_2 = \frac{k^2}{4} J_k(k),$$
$$a_3 = 0, \qquad a_4 = -ka_1, \qquad a_5 = 2ka_2, \qquad a_6 = -6ka_1.$$

Here J_k are Bessel functions. In general, the coefficients a_j in (45) satisfy recurrence relations of order 3 (Sadov 2006, (42))

$$a_{j+3} = j(j-1)a_{j+1} + kj(2j-3)a_j. \tag{46}$$

Example 3 In this example we demonstrate one application of the identities (27) and (24). A general result—Proposition 6—is then specialized to cover the functions of Example 2 with subscript $k < 0$.

Proposition 6

(a) *If $p \geq 0$ and $q \geq 0$, then $F^{p,q}_{r,\tau}(\eta) = (-1)^{p+1} F^{p,q}_{\tilde{r},\tilde{\tau}}(-\eta)$.*

(b) *If $p \geq 0$ or $q \geq 0$, then $F^{p,q}_{r,\tau}(0) = (-1)^{p+1} F^{p,q}_{\tilde{r},\tilde{\tau}}(0)$ and*

$$F^{p,q}_{r,\tau}(\eta) = (-1)^{p+1} F^{p,q}_{\tilde{r},\tilde{\tau}}(\eta) + P(\eta)(\eta+1)^{-\nu_1}(\eta-1)^{-\nu_2} \exp(\tau\eta), \tag{47}$$

where $P(\eta)$ is a polynomial depending on p, q, r, τ and $\nu_{1,2} \geq 0$ are some integers.

Proof If $p \geq 0$ *and* $q \geq 0$, then ${}_1F =_\xi F = 0$ and the result (a) follows from (27). If $p \geq 0$ *or* $q \geq 0$, then according to Proposition 4(a) and reciprocity (24), the extra term ${}_1F(-\eta) - {}_\xi F(-\eta)$ in the right-hand side of (27) is analytic at $\eta = 0$. But $\xi = 1$ when $\eta = 0$, so the difference is 0, which proves the first part of (b). For general η, we have by (24) and Proposition 4(a)

$${}_\xi F^{p,q}_{\tilde{r},\tilde{\tau}}(-\eta) =_1 F^{q,p}_{r,\tau}(-\eta) R_1(\eta) = \exp(\tau\eta)\, R_2(\eta),$$

where R_1, R_2 are rational functions with possible poles at $\eta = \pm 1$. Hence $R_2(\eta) = P(\eta)(\eta+1)^{-\nu_1}(\eta-1)^{-\nu_2}$, $\nu_{1,2} \geq 0$.

For the Hansen coefficients of Example 2, the rational function $R_2(\eta)$ is computed in the explicit form (Sadov 2006, Theorem 5(b)):

$$F^{0,-4}_{1-k,k}(\eta) = F^{0,-4}_{1+k,-k}(-\eta) - \frac{k}{6}\frac{(1-\eta)^{k-2}}{(\eta+1)^{k+2}} \exp(k\eta).$$

Consequently, the following relation holds (Sadov 2006, Theorem 6(c)):

$$X^{-3,2}_k|_{\eta=0} = X^{-3,2}_{-k}|_{\eta=0} - \frac{2k}{3}.$$

9 Conclusion

The main results of the paper are stated in three theorems. Theorem 1 asserts that Hansen's coefficient $X_k^{n,m}$ with even $m-k$ is a meromorphic function of the variable $\eta = \sqrt{1-e^2}$ with possible poles only at $\eta = 0, \pm 1$. Lower estimates for orders (of poles or zeros) at these special points are obtained; the result for $\eta = 1$ is consistent with the classical D'Alembert characteristic of the Maclaurin series of $X_k^{n,m}$ in powers of e (Remark 2 in Sect. 7). If $m-k$ is odd, the above statements hold true for the function $(1-\eta^2)^{1/2} X_k^{n,m}(\eta)$. Factoring away the poles yields an entire function of η with exponential growth estimates (Theorem 2). The entire function of η can be split into the odd and even parts and thus expressed in terms of two entire functions of e^2 (Theorem 3).

On a technical level, a residue decomposition of the contour integral that defines the nontrivial factor in the multiplicative decomposition of $X_k^{n,m}$ is studied (Sect. 3). A more general family of functions depending on four parameters, from which Hansen's coefficients are derived by specialization, is considered. Two involutive transformations of variables (Sect. 4) are employed to derive useful relations between the members of the family.

The qualitative results stated in Theorems 2 and 3 provide, in particular, for a possibility to construct rapidly convergent series in any of the two parameters, η or e, for evaluation of $X_k^{n,m}$ uniformly on the interval $[0, 1]$. The word "rapidly" refers to the rate of decay of Taylor coefficients when n, m, and k are fixed. We did not attempt to address practical computational issues arising due to non-uniformness of the estimate (42) in n, m, and k. Also, algorithmic aspects of evaluation of the coefficients are beyond the scope of this paper.

Acknowledgements I am indebted to Prof. V.V. Sidorenko for pointing out the connection between the Chernousko functions in Sadov (2006) and Hansen's coefficients. A financial support by a grant from NSERC the Natural Sciences and Engineering Research Council of Canada, is acknowledged.

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